

Project Initialization and Planning Phase

Date	June 26, 2026
Team ID	SWTID-2026-6095
Project Title	EduGenie - AI-Powered Educational Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform educational support using generative AI, boosting accessibility, personalization, and learning efficiency. It tackles the fragmentation of existing study tools, promising faster concept understanding, better self-assessment, and more engaged learners. Key features include an AI-powered Q&A and explanation engine, automated quiz generation, text summarization, and personalized learning path recommendations.

Project Overview	
Objective	The primary objective is to simplify and enhance the learning experience by providing instant, AI-powered educational support – helping learners understand concepts, answer questions, generate quizzes, summarize material, and receive personalized learning paths.
Scope	The project delivers a lightweight, unified educational assistant that combines Q&A, concept explanation, quiz generation, summarization, and learning path recommendations into a single FastAPI-powered application, accessible to students, self-learners, and educators across academic levels and subject domains.
Problem Statement	
Description	Learners currently rely on multiple disconnected tools – search engines, video tutorials, generic chatbots, and private tutoring – to get explanations, self-assess understanding, summarize material, and plan their studies, which is inefficient, time-consuming, and often inaccessible on resource-constrained devices.
Impact	Solving this will result in faster concept mastery, improved self-assessment through instant quizzes, reduced study time via summarization, and clearer, personalized learning progression – ultimately improving learning outcomes and learner confidence.
Proposed Solution	
Approach	Building a lightweight AI-powered educational assistant using FastAPI (backend) and an HTML/CSS frontend, combining lightweight local models with cloud-based generative AI (Gemini AI) to deliver fast, accurate, and interactive educational support.
Key Features	- Intelligent Question Answering for instant, accurate responses - Simplified Concept Explanations for complex topics - AI-Powered Quiz Generation for self-evaluation - Educational Text Summarization of learning material - Personalized Learning Path Recommendations (beginner to advanced) - Optimized to run efficiently on resource-constrained devices such as the Mac M1

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	Apple M1 (8-core CPU) or equivalent – no dedicated GPU required
Memory	RAM specifications	8 GB
Storage	Disk space for app, models, and logs	256 GB SSD
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn
Libraries	Additional libraries	google-generativeai (Gemini AI SDK), pydantic, python-dotenv, requests, Jinja2
Development Environment	IDE	VS Code, Jupyter Notebook
Data		
Data	Source, size, format	Sample educational Q&A and quiz datasets (Kaggle/open educational sources); user-provided study text for summarization – variable size, CSV/JSON/plain text